

Noah Li

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Education

University of Wisconsin-Madison

May '27

- **GPA:** 3.8/4.0
- **B.S.** in Computer Science, **B.S.** in Data Science
- **Relevant Coursework:** Algorithms, Matrix Methods in Machine Learning, Machine Learning and Statistical Pattern Classification, Data Structures (Java), Database Management, Big Data Systems, Data Science Modeling I & II, Discrete Mathematics, Machine Organization and Architecture, Matrix and Linear Algebra

Experience

Caterpillar

May '25 – Present

Software Engineer Intern

- Developing Cloud-native infrastructure and distributed systems; focusing on migrating high-volume data streams to high-availability cloud environments to support predictive AI modeling in Data & AI Helios Team
- Designed and deployed a scalable, end-to-end REST API to streamline real-time transfer of manufacturing and tooling issue data across cross-functional teams, enhancing operational visibility and reducing resolution latency.
- Engineered and automated OEE (Overall Equipment Effectiveness) monitoring pipelines using Python, leveraging data-driven insights to significantly optimize operator workflows and drive continuous improvement initiatives on the manufacturing floor.
- Spearheaded a cross-functional engineering team to design and deliver a C/Python application for statistical process capability (CPK) analysis of fuel injectors, ensuring rigorous quality control and supporting defect reduction

Data Science For Sustainable Development

Jan '25 – Present

Software Engineer Lead

- Architected a digital archive platform for George Mason University using Go and HTMX, leveraging AWS S3 for optimized storage
- Integrated Amazon Bedrock API to implement LLM-driven features for previously inaccessible historical archives
- Leveraged AWS EC2 to host the Go/HTMX engine, ensuring low-latency integration with Amazon Bedrock

Komatsu

Jun '24 – Aug '24

Data Engineer Intern

- Migrated and digitized 15,000+ Fortran Material Specifications into Excel, creating a centralized database accessible to all Komatsu Engineering teams
- Performed comprehensive data validation and quality assurance for engineering documentation across multiple departments
- Streamlined information access across Admin and Engineering groups, reducing search time by 20% and enhancing cross-departmental collaboration

Projects

Re-LLM

- Built an end-to-end Reinforcement Learning through Human Feedback training pipeline on a 1-3B Large-Language Model using HuggingFace TRL, replicating the core alignment technique behind other popular LLMs
- Fine-tuned through the full RLHF pipeline: supervised fine-tuning on demonstration data, reward model training on human preference pairs, and PPO with a KL penalty to prevent the model from gaming its own reward signal
- Benchmarked PPO against DPO as a simpler alternative; experimented with the KL coefficient β to explore the tradeoff between alignment and model capability

Efficiency Navigator

- Led a student team building a React + Vite + Leaflet web map for Madison nonprofit Sustain Dane that ingests a multi-sheet Excel workbook as part of their Efficiency Navigator Program, making housing more affordable and resilient in the Dane County area
- Visualizes energy, CO₂, VMT, and dollar-savings metrics for 200+ cross-street intersections in the Efficiency Navigator Program
- Drove UX and data-visualization requirements with zoom-reactive SVG markers, address-to-nearest-intersection search, geolocation auto-zoom, Recharts donut metrics, and plain-language explainer cards, using AGILE Methodologies

MadGym

- Developed a web app that shows live occupancy for campus recreation zones (e.g. fitness floors, courts, pool) so users can see how busy each area is before going.

Skills

- **Languages:** C, C++, C#, Java, Python, GoLang, SQL, TypeScript, HTML/CSS, JavaScript, R
- **Frameworks & Tools:** AWS, Docker, Linux/UNIX, React, Git, Jupyter, LaTeX, Beautiful Soup, Snowflake, PowerBI, Ignition, Tableau
- **Libraries:** Pandas, NumPy, Selenium, Matplotlib, XGBoost, Leaflet